Labels Correct by Construction May Be Unidentifiable from Observation: Training a VLM for Mathematical Error Diagnosis on Synthetic Handwriting-Style Answer Sheets

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Abstract

Training a vision–language model (VLM) to diagnose errors in handwritten mathematics requires answer-sheet images annotated with error locations, error types, and scores—fine-grained labels that are practically unavailable for real answer sheets. We build a *label-by-construction* pipeline: solutions are generated as executable programs, errors are injected as verified program mutations, and deterministic rendering makes every character coordinate known. Supervised fine-tuning on 9,204 synthetic samples establishes error localization that never emerges zero-shot (IoU@0.5 0.000 $\rightarrow$ 0.795) and reduces hallucinated errors from 31.4% to 0%. We then quantify, by exhaustively scanning the generator's output, a failure mode specific to synthetic data: labels that are correct by construction can be *unidentifiable from observation*. 95.6% of "conceptual misconception" labels are string-identical on the answer surface to "mechanical slip" labels, and the unmet type-accuracy target (0.760) matches a conditional expectation under majority-side tie-breaking (0.759). Mapping each ground-truth label to the *finest granularity identifiable from the observation* and penalizing finer-grained assertions on unidentifiable cases as *over-claiming* yields type accuracy 0.970 with 0% over-claiming *via inference-time label mapping alone, without retraining*; on a freshly generated evaluation set, over-claiming drops from 90% to 1%.

1 Introduction

Automated grading of written mathematics requires diagnosing *where* in the working an error occurred and *of what kind*, not merely whether the final answer is correct. Images of real answer sheets exist in abundance, but training data that jointly provides error coordinates, fine-grained error types, propagation flags, and error-free control images does not, to our knowledge, exist in usable form. Manual annotation at 10 minutes per page would cost roughly 3.3M per 10,000 pages. We resolve this supply problem by *label by construction*: solutions are generated as executable step sequences, errors are injected as verifiable program mutations, and rendering makes all character coordinates known (Fig. 1). Labels are derived from the construction rather than annotated.

Our contributions are threefold.

1. **Pipeline.** An implementation comprising solution-program mutation, two gates (execution verification / diff containment), *composition-based verbalization* (the LLM produces only connective explanation clauses; equations are composed mechanically from templates, so equation fidelity is guaranteed by construction), and pre-training adversarial testing of the reward function. SFT on 9,204 samples establishes localization and typing abilities that are exactly zero in zero-shot evaluation (§4).

2. **Quantifying and attributing a failure mode.** We show, by exhaustive scanning, that construction-correct labels can be unidentifiable from observation (observational-equivalence rate $\rho = 0.956$). Diagnostic ambiguity is classical (§2), but for synthetic data the *amount* of ambiguity is exactly computable from the generator; we attribute the failure to a design error—*using generation provenance as the prediction target*—and show it is deterministically fixable (§5).

3. **Prescription.** Map each ground truth to the finest observation-identifiable label and penalize finer assertions on unidentifiable cases as *over-claiming.* Inference-time label mapping alone—no retraining—achieves type accuracy 0.970 with 0% over-claiming. "Saying it cannot be determined when it cannot" matches the behavior of a pedagogically desirable grader that does not assert "conceptual misconception" without evidence.

2 Related Work

Synthetic document data. Rendering-based training data is standard in document understanding and OCR (SynthText [6]; SynthDoG in Donut [7]), but supplies transcription supervision. We synthesize counterfactual fine-grained labels: error location, type, and scoring. **Mathematical error diagnosis.** That a single erroneous surface can arise from multiple procedural bugs—diagnostic ambiguity—has been known since BUGGY [1]. Recent work addresses erroneous-step identification in textual reasoning [8, 9] and VLM evaluation on handwritten mathematics (FERMAT [10]; CHECK-MAT [11]), but not the *supply of coordinate-level fine-grained training labels.* JaWildText [12] benchmarks Japanese scene-text understanding including handwriting OCR. **Backing off to coarser granularity; partial labels.** Penalizing over-specific predictions on a hierarchy is proposed by Deng et al. [4]; labels as sets of observationally equivalent candidates correspond to partial-label learning [3]; selective prediction [2, 5] handles abstention along the confidence axis. **Our novelty is not these concepts** but the exact, exhaustive quantification of the ambiguity inside a synthetic-data generator—an amount that must be estimated for annotated data—and its deterministic repair by relabeling. **Quality of synthetic data.** Surveys cover factuality and fidelity [14]; we are not aware of prior measurement and repair of the "correct by construction yet observationally unidentifiable" failure mode.

3 Synthetic Pipeline

Fig. 1 gives an overview. **M1**: problems and reference solutions in two domains of grade-7 mathematics (signed arithmetic; linear equations) are generated as executable step sequences; execution verifies all intermediate values (gate G1a). **M2**: errors are injected as program mutations—mechanical operators (transposition-sign, dropped product sign, arithmetic slips) and LLM *concept operators* that propose misconception-based wrong expressions, validated by an AST whitelist parser. Re-execution and diff containment (invariance outside the mutation site and its causal descendants) are guaranteed by gate G2. Error-free controls make up 30%. *Twin pairs* share the surface text of all steps preceding the mutation site (100% agreement across all 3,000 pairs); only causally affected steps are verbalized independently. **Composition-based verbalization**: the LLM outputs only explanation clauses; equations are composed mechanically from step templates (gate G1b), so equation fidelity is guaranteed by construction (a prompt-instruction variant passed only 8–16/20 across runs and was rejected; the composition method passed a 40/40 smoke test). The verbalization pass rate was about 95% over 10,000 samples. **Pre-training verification**: adversarial testing of the reward function (1,000 malicious outputs; final inversion 0/1000) caught

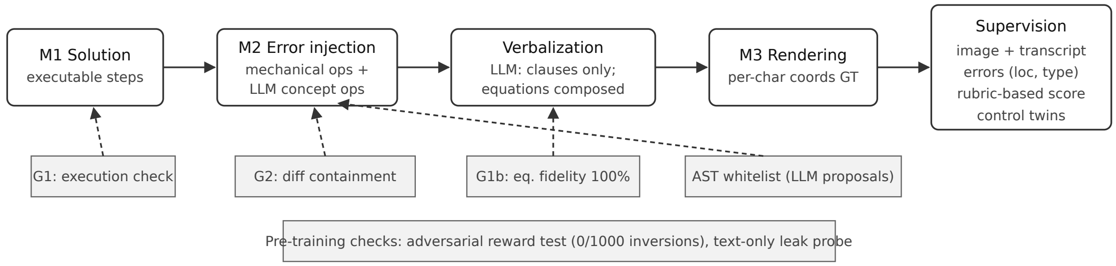

Figure 1: The label-by-construction pipeline. Labels are derived from the construction, not annotated.

Table 1: Zero-shot → SFT (2B, llama.cpp Q8, n=200 = 149 errors / 51 controls).

Metric	Zero-shot	After SFT
Localization IoU@0.5	0.000	**0.795**
Type accuracy (9-class macro)	0.000	0.647
Detection TPR	1.000	0.980
Detection TNR	0.686	**1.000**
Hallucination rate	0.314	**0.000**
Score exact match	0.345	**0.935**

three reward bugs before any training. A character-bigram probe predicting error presence from the *reference transcription text only* (group split by pair) served as a leak diagnostic; AUC 0.63 under the old generator (integer-solution constraint) fell to 0.53 after allowing fractional solutions (a diagnostic, not a proof of absence).

Scale: 10,000 samples (7,000 errors / 3,000 controls) generated in about 7 hours with full rendering success, using Qwen3.6-35B-A3B (4-bit, llama.cpp) on a single mini-PC.

4 Experiment 1: Zero-Shot to SFT

Metrics. IoU@0.5: fraction of ground-truth errors whose predicted bbox reaches IoU $\geq$ 0.5. Type accuracy (macro): class-wise accuracy averaged over classes. Hallucination rate: fraction of control sheets with at least one reported error. Score exact match: fraction of sheets with exactly matching total score. Over-claiming rate: among errors whose ground truth is family-granularity (unidentifiable), the fraction where the model asserts a subtype (§5).

Setup. Qwen3-VL-2B-Instruct, QLoRA (4-bit; language layers only; r=16, α=16, lr 2×10^{-4}, effective batch 8, 1 epoch), trained on **9,204** samples (pair-wise split 9,204/489/307 of 10,000; one TITAN V; about 2.5 h). Input: sheet image + problem + reference solution + rubric; output: JSON (faithful transcription, error list with step ID / relative bbox / type, score). Evaluation: 200 held-out samples *isolated pair-wise from the same generation pool* (no instance overlap; distribution shared), measured in deployment form (GGUF 8-bit + llama.cpp, no system prompt, temperature 0); see Table 1.

The zero-shot column uses a system prompt and absolute-coordinate contract; the SFT column uses no system prompt and relative coordinates, so the comparison is not fully controlled (an ablation showed the system-prompt mismatch alone moves score exact match 0.775 → 0.935). Zero-shot

Table 2: Type accuracy on the same instrument (held-out 200).

Configuration	Type acc.
Old labels, old 9-class eval	0.7595
Old labels + inference-time mapping (no retraining)	**0.970**
Retrained on identifiable labels (3 seeds)	0.976±0.010

Table 3: Freshly generated evaluation set (507 samples, llama.cpp Q8; the retrained model has one parse failure; 20 cross-family-ambiguous samples are excluded from type metrics).

Metric	Old labels	Identifiable
Over-claiming rate	0.900	**0.010**
Family accuracy (7-class avg.)	0.31	0.806
Type accuracy (9-class macro)	0.596	0.627
Detection TPR	**0.968**	0.874
Localization IoU@0.5	**0.794**	0.756
Score exact match	**0.909**	0.840
TNR / hallucination	1.000/0%	1.000/0%

“detection” is the degenerate report-everything behavior (TPR 1.000 / TNR 0.686). Localization and typing are exactly zero for 2B/4B/8B zero-shot; SFT establishes them. Hallucination reaches 0% after SFT with control twins (0/51 controls; the 95% CI upper bound is about 7%; the isolated contribution of control pairs is not ablated).

5 Experiment 2: Label Identifiability

Finding. Type accuracy missed its 0.85 target; confusions concentrated within subtype pairs (e.g., *conceptual transposition misunderstanding* vs. *notational transposition-sign slip*) and their direction flipped across runtimes. Define the **observational-equivalence rate** ρ as the fraction of one operator's outputs whose answer-sheet surface (string) coincides with another operator's output. Exhaustive scanning showed: (i) $\rho = 731/765 = 0.956$ for the conceptual transposition operator (Fig. 2); (ii) misconception-verbalizing clauses occur in 0/765 samples (the generator never passes the misconception rationale to the verbalizer); (iii) for the product pair, 84.2% of the coexistence region ($q{\cdot}r < 0$) is observationally equivalent, and the seemingly identifiable remainder is a *spurious identifiability region* where the competing operator is structurally never eligible. Subtype labels are thus *records of the generation path*, not observable properties of the sheet; using them as prediction targets is the root failure (label-by-construction itself is intact). As a consistency check, a conditional expectation assuming majority-side tie-breaking and measured performance on non-colliding classes gives 0.7593, close to the measured 0.7595 (difference 0.0002). The analysis is also a negative result for the LLM concept operator: semantic diversity of mutations is worthless unless the mutation leaves a differential trace in the observation.

Fix. Map each type ground truth to the finest label identifiable from its observation (family for observationally equivalent cases; subtypes retained only for the identifiable minority—1.5% of all errors). Evaluation and reward become three-valued: exact match at identifiable granularity = correct; family-only answer where a subtype is identifiable = partial credit; **subtype assertion where only the family is identifiable = incorrect (over-claiming)**. The mapping is a deterministic predicate over existing data; no regeneration is required.

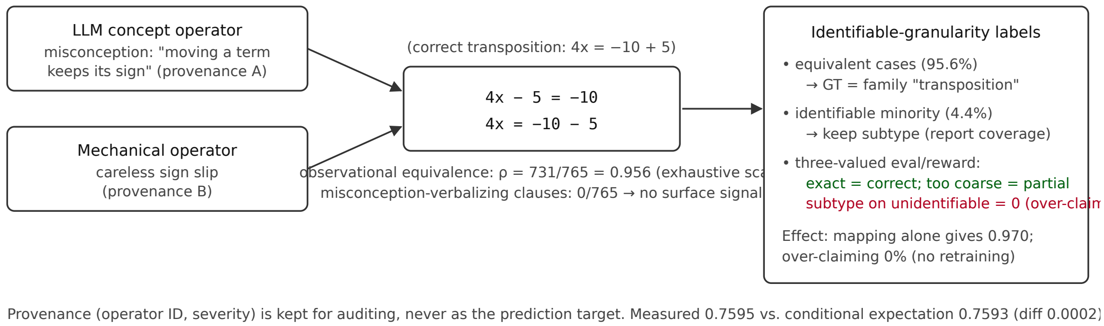

Figure 2: Observational equivalence and identifiable-granularity labels (running example shared with Fig. 3).

Retraining is unnecessary (Table 2). The inference-time mapping produces no over-claiming by definition (0%) and yields type accuracy 0.970. Retraining on identifiable labels suppresses over-claiming natively to 1% (Table 3) but shows no additional type-accuracy gain; the retained subtypes (1.1% training coverage) are not learned (0/17 on the fresh set), and TPR, IoU, and score regress on the fresh set (single runs each; variance not separated). The retrained model's family average (0.806) is below our acceptance floor (0.90; under investigation—the held-out set shares the training generation pool, which is the leading explanation for the gap to 0.964). **The practical recommendation is therefore the old model plus inference-time mapping.**

6 Limitations

(1) All results are within synthetic handwriting-style rendering; transfer to real handwriting awaits a planned glyph-bank collection. (2) Retraining was repeated over three seeds (primary artifacts in the repository), but detection/score/transcription sit at ceiling, so zero variance there is not evidence of robustness; fresh-set comparisons are single runs. (3) Causally affected steps of twin pairs are verbalized independently; pre-mutation steps agree in 3,000/3,000 pairs and a whitespace-style probe scores AUC 0.530 (chance), but this is not an exhaustive leak audit. (4) The teacher-agreement study (4 graders $\times$ 100 shared sheets) is still being arranged; human validity of the label design is unverified. (5) Experiments cover two domains of grade-7 mathematics, one generator, one model (2B); quantitative conclusions are limited accordingly. What generalizes is the *audit procedure*: (a) measure the observational-equivalence rate ρ between generation paths by exhaustively scanning the generator's output; (b) map high-ρ distinctions out of the prediction target to the finest observable granularity; (c) penalize finer assertions on unidentifiable cases. That ρ is exactly computable is an advantage unique to synthetic data.

7 Conclusion

We synthesized answer sheets with fine-grained counterfactual labels by construction and established localization and scoring in a 2B VLM. We further quantified, by exhaustive generator scanning, that construction-correct labels can be observationally unidentifiable, and showed that identifiable-granularity mapping with an over-claiming penalty removes pedagogically harmful unfounded assertions *without retraining*. Code is available in the public repository.

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A A Generated Example

Fig. 3 shows a real generated pair (problem $4x-5=-10$; conceptual transposition misconception injected at step s2; score ground truth 3/10). The injection changes a single character ("+5"→"−5"), and the mechanical operator produces the identical surface—the observational-equivalence instance of Fig. 2. The sheets are in Japanese, as generated by the pipeline.

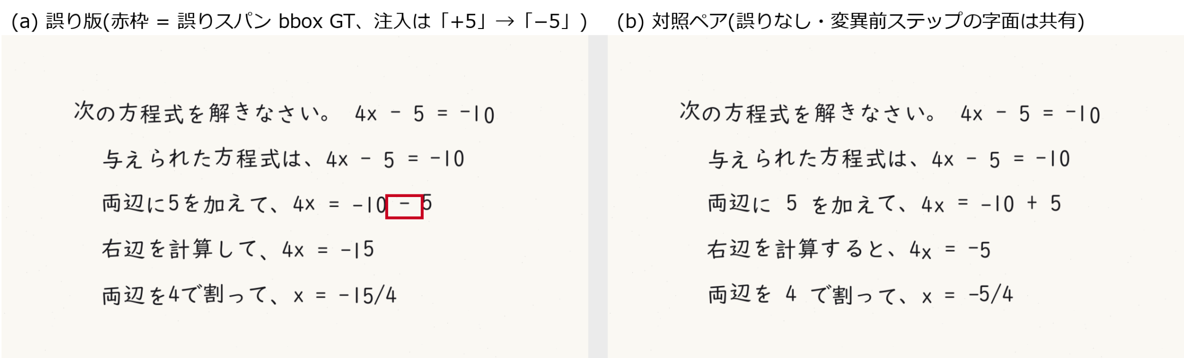

Figure 3: A generated example: (a) erroneous sheet (red box = bbox ground truth), (b) error-free control twin sharing pre-mutation surface text.